

Algorithms Problem Solving

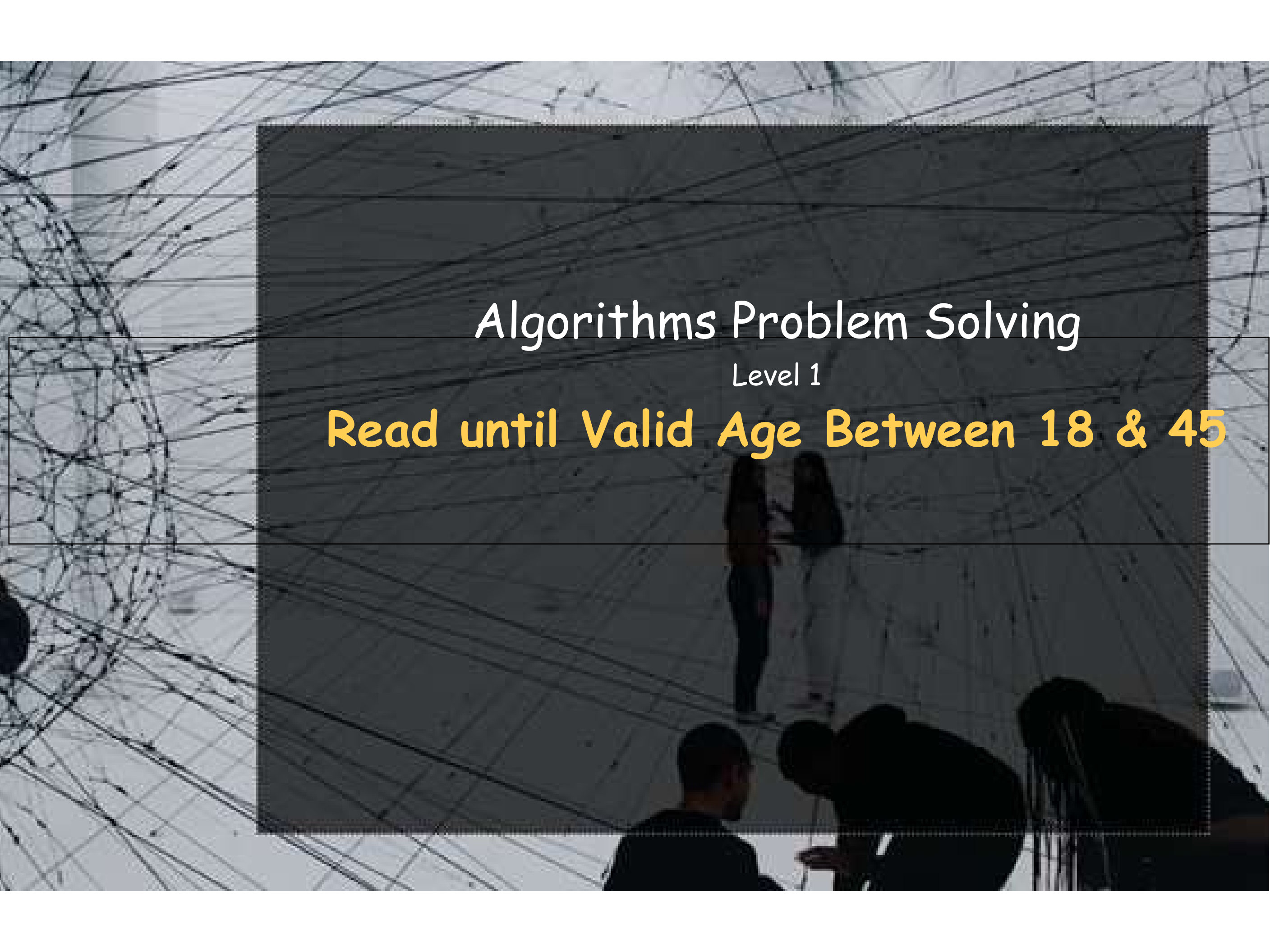
Level 1

Read Until Valid Age
Between 18 & 45

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Read until Valid Age Between 18 & 45



Definition

- ◆ This program repeatedly asks the user to enter an age until a valid age between 18 and 45 is entered.
- ◆ The program does not continue until the input satisfies the required range.
- ◆ It uses a `do-while` loop together with a Boolean validation function.

Explanation

- ◆ ``readAge()`` reads the user's age and returns it.
- ◆ ``validateNumberInRange()`` checks whether the number is between ``From`` and ``To``.
- ◆ ``readUntilAgeBetween()`` keeps reading the age until the validation function returns ``true``.
- ◆ The ``do-while`` loop guarantees that ``readAge()`` executes at least once.
- ◆ ``!validateNumberInRange(...)`` means "the age is not valid".
- ◆ As long as the age is invalid, the loop continues.
- ◆ When a valid age is entered, the condition becomes false and the loop stops.
- ◆ ``printResult()`` then prints the valid age.
- ◆ ``readUntilAgeBetween(18, 45)`` passes 18 as the minimum and 45 as the maximum.

Steps in main()

1. Set the minimum age to 18 and maximum age to 45.
2. Read the user's age.
3. Check whether the age is between the two limits.
4. If the age is invalid, repeat the input.
5. Continue until a valid age is entered.
6. Return the valid age.
7. Print the valid age.

💡 Example

◆ If the user enters:

Age: 15

Age: 50

Age: 25

◆ 15 is invalid, so the program asks again.

◆ 50 is invalid, so the program asks again.

◆ 25 is valid, so the loop stops.

➡ Result

Your Age is: 25

Note

- ◆ The `!` operator reverses a Boolean value:
 - `true` → `false`
 - `false` → `true`
- ◆ Therefore:
 - `!validateNumberInRange(age, From, To)`
- ◆ means the loop continues while the age is NOT valid.

⊘ Warning

- ◆ The program prevents an out-of-range age from being accepted, but it does not handle non-numeric input such as letters.



Comparison

`do-while`

- ✓ Executes the input at least once.
- ✓ Checks the condition after the loop body.
- ✓ Suitable when input must be requested at least once.

`while`

- ✓ Checks the condition before executing the loop body.
- ✓ The body might execute zero times.
- ✓ Useful when the condition is already known before the first iteration.
- ✓ At least one condition is false.

✦ Summary

- ✓ `do-while` is useful for repeated input validation.
- ✓ The loop continues while the entered age is invalid.
- ✓ `!` reverses the Boolean result.
- ✓ The user must enter an age between 18 and 45 before the program continues.
- ✓ The valid age is returned from `readUntilAgeBetween()` and passed to `printResult()`.

Thank you 😊